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Research on the Construction of Ability Portrait of Highly Skilled Talents Based on Knowledge Graph and AI Personalized Training Path Planning

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Abstract

As a national basic industry, the cultivation of high-skilled talents in the electric power industry is of great significance to guarantee the safety of power grids and promote technological innovation. Aiming at the problems of insufficient dynamic adaptability of high-skilled personnel's ability and job requirements in the digital transformation of the electric power industry, and the inefficiency of traditional training "one-size-fits-all", this study proposes a competency portrait construction method based on the fusion of multi-source data and an AI-driven personalized training path planning model. Firstly, structured and unstructured talent data are acquired through multi-source data acquisition and preprocessing methods, and cleaning and standardization are carried out to provide data support for subsequent modeling; secondly, combining with the job characteristics of highly qualified personnel in the electric power industry (e.g., new energy operation and maintenance and smart grid overhauling), the competency portrait model of highly skilled personnel in the electric power industry is constructed based on the knowledge mapping technology to realize the following Secondly, based on knowledge mapping technology, we build a competency portrait model of highly skilled talents in the electric power industry based on knowledge mapping technology to realize a three-dimensional competency portrait framework of "skill-knowledge-competency". Take the training team of a provincial power company as an example for experimental verification. The results show that the method can improve the matching degree of training content by 33% and shorten the employee skill attainment cycle by 25%, providing technical support for the accurate training of enterprise talents.

CCS Concepts

• **Computing methodologies** → Machine learning; Machine learning algorithms.

Keywords

multi-source data, competency profiling, knowledge mapping, deep learning, reinforcement learning, personalized training

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1 Introduction

1.1 Background and significance of the study

With the promotion of the "dual carbon" goal and the construction of new power systems, the power industry is undergoing a rapid transformation from traditional thermal power to new energy (wind power, photovoltaic), smart grid, and energy storage technology. Highly skilled personnel (such as new energy power station operation and maintenance engineers, intelligent substation commissioning technicians), as the core carrier of technology implementation, their ability level directly affects the safety and stability of the power system and efficiency improvement. However, the current cultivation of highly skilled personnel in the power industry faces two major pain points:

(1) Dynamic changes in competency needs: new energy equipment (such as photovoltaic inverters, energy storage PCS) technology iteration is fast, and the traditional experience-based job competency standards are difficult to update in real time.

(2) mismatch between supply and demand for training: corporate training relies on the "standardized curriculum + centralized lectures" model, ignoring the differences in the individual skills of employees (such as hands-on experience, learning preferences), resulting in a waste of training resources and skills enhancement effect is insufficient (according to the China Federation of Electricity Enterprises 2024 research, about 41% of the front-line employees believe that the training content and the actual work demand deviation). (according to the research of China Electricity Council in 2024, about 41% of frontline employees believe that the training content deviates from the actual work demand).

In this context, combining artificial intelligence (AI) technology to build a competency portrait of highly skilled personnel, and realizing "competency-oriented" personalized training path planning has become a key direction for the transformation of human resource management in the power industry. Competency profiling accurately portrays the current status of employee competencies through multi-dimensional data mining, providing a basis for training content customization; The AI-driven path planning can dynamically adjust the training strategy according to the competency gaps,



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and finally realize the efficient matching of "person-post-capability" [1].

1.2 Research status and innovations

In existing research, competency portrait technology has been applied in manufacturing (e.g., intelligent manufacturing) and medical industry (e.g., clinician competency assessment), but the power industry has not yet formed a mature competency portrait framework due to the complexity of the job scenarios (involving multi-dimensional aspects such as equipment operation and maintenance, dispatching and control, and safety management), and the variety of data types (including structured assessment records, unstructured operation logs, and implicit experiential knowledge). In the field of personalized training, traditional methods are mostly based on rule matching or simple collaborative filtering algorithms, which are difficult to adapt to the dynamics of electric power skills and the personalization of employee needs.

The innovations of this study are:

(1) Propose a three-dimensional "skill-knowledge-competence" framework for profiling the competence of highly skilled electric power personnel, which integrates heterogeneous data from multiple sources, such as job task logs, practical behavior data, and assessment results.

(2) Design a competency labeling system based on knowledge mapping to solve the problem of visualization of hidden experience in the field of electric power.

(3) Construct a personalized training path planning model driven by reinforcement learning to achieve dynamic and optimal allocation of training resources.

2 Knowledge Graph-Based Model for Building Competency Profiles of Highly Skilled Talent

2.1 Multi-source data collection and preprocessing methods

In the cultivation system of highly skilled personnel in the electric power industry, the collection and preprocessing of multi-source data is a key preprocessing link in the construction of a competency portrait. With the increasing complexity of the electric power system, the traditional training methods can hardly meet the demand for fine identification and personalized path planning of talent's ability. For this reason, researchers have begun to focus on how to integrate data resources from different channels, including unstructured and structured data such as operation logs of simulation training platforms, skill assessment results, learning behavior trajectories, and experts' experiences, to build a comprehensive and dynamic competency portrait system. In the data collection phase, API interfaces, database extraction, log parsing, and natural language processing technologies are comprehensively applied to unify and aggregate heterogeneous data sources such as power simulation systems, online learning platforms, and records of practical training centers. In the data preprocessing stage, key steps are involved in missing value filling, outlier detection, data standardization, and feature coding to ensure the accuracy and stability of subsequent modeling^[2]. To cope with the problem of variable data quality, an automatic cleaning mechanism based on deep learning

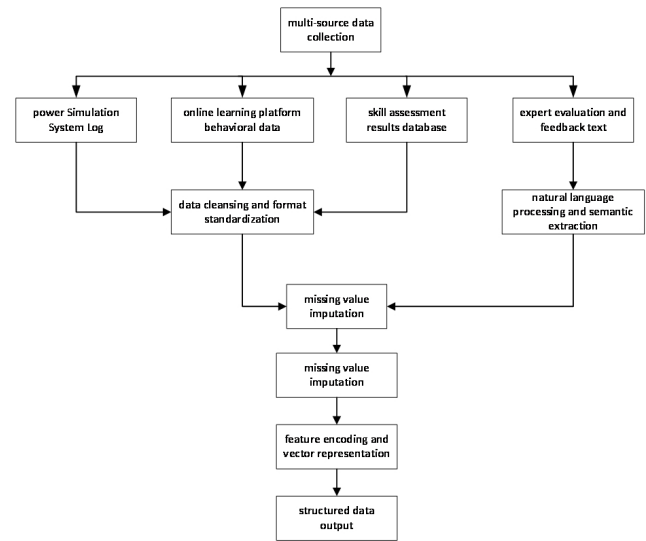


Figure 1: The overall process framework of multi-source data acquisition and preprocessing.

is introduced in the study, which identifies and corrects the noise and redundant information in the data through the structure of the self-encoder network to improve the data usability^[3]. The overall process framework of multi-source data acquisition and preprocessing is shown in Figure 1 below.

2.2 Knowledge graph-based competency portrait modeling

With the rapid development of the electric power industry, the ability portrait modeling of highly skilled personnel has become a key link to improve the efficiency of human resource allocation and training quality. With the continuous maturation of artificial intelligence technology, especially the integration of knowledge graph and deep learning, it provides technical support for building accurate, dynamic and scalable ability portraits. By integrating multi-dimensional data such as power system operation, equipment maintenance, safety management, etc., competency portraits can not only reflect the current status of individual skill levels, but also predict the evolution path of their competencies, providing a basis for subsequent personalized training^[4].

Knowledge mapping, as a semantic network tool connecting structured and unstructured data, can extract entities and map relationships between job responsibilities, skill standards, training resources, equipment operating procedures, and other information in the electric power industry, to construct a competency knowledge base with industry characteristics. The knowledge base not only includes explicit knowledge (e.g., operating procedures, equipment parameters), but also covers implicit experience (e.g., fault judgment logic, emergency response strategy), thus forming a competency model framework with semantic reasoning ability. For example, when constructing the competency portrait of substation operation and maintenance personnel, knowledge mapping can correlate

Table 1: Typical capability dimensions represented in knowledge graphs.

Competency dimensions	Knowledge Graph Entity Type	Data source	Assessment method	Weighting factor	Update Frequency
Equipment operation proficiency	Equipment, operating procedures	Training records, simulation system	Operation hours, accuracy	0.25	Quarterly
Troubleshooting ability	Troubleshooting cases, diagnostic rules	Work order system, expert experience	Diagnosis accuracy, response time	0.30	Real-time
Awareness of safety regulations	Safety regulations, accident cases	Safety training, examination results	Pass rate, violation records	0.15	Semi-annually
Technical documentation comprehension	Documentation, definition of terms	Learning platform, knowledge base	Reading hours, test scores	0.10	Monthly
Teamwork skills	Projects, Roles	Project management system, evaluation forms	Task completion, feedback ratings	0.20	Per project cycle

the "GIS combination appliance operation procedure" with the "abnormal signal recognition ability", and then explore the potential competency dependencies through graph neural networks^[4].

On this basis, when combining heterogeneous data from multiple sources for portrait modeling, deep learning-based embedding representations, such as graph convolutional networks or Transformer architectures, are usually used to map high-dimensional capability features into low-dimensional vector spaces. By introducing the attention mechanism, the model can automatically identify the importance weights of different ability dimensions, and then realize the dynamic assessment of individual ability status, whose mathematical expression can be expressed as Equation 1).

$$h_i^{(l+1)} = \sigma \left(\sum_{j \in N(i)} \alpha_{ij} W^{(l)} h_j^{(l)} \right) \quad (1)$$

Where $h_i^{(l)}$ denotes the embedding vector of the i th node in the l th layer, $N(i)$ denotes the set of its neighboring nodes, α_{ij} denotes the attention weight, $W^{(l)}$ is the learnable parameter matrix, and σ is the activation function. This formula describes the process of information aggregation and feature updating in the knowledge graph, which helps to improve the accuracy and generalization of capability portraits^[5].

To further illustrate the key elements involved in competency portrait modeling, Table 1 below lists typical competency dimensions and their representation in the knowledge graph:

Through the above method, the knowledge graph-driven competency portrait not only realizes the structured modeling of the competencies of highly skilled personnel in the electric power industry, but also provides a solid foundation for the subsequent personalized training path planning based on reinforcement learning.

3 AI-based personalized training path planning with integrated competency profiling

3.1 Application of deep learning in training demand analysis

In the cultivation of highly skilled personnel in the electric power industry, the accurate identification of individual training needs is a key link in improving training effectiveness. The traditional training demand analysis method usually relies on questionnaire surveys or expert assessment, which cannot meet the large-scale and dynamically changing talent ability development needs. In recent years, the rapid development of deep learning technology has provides a new technical path for building intelligent and personalized training demand analysis models. By constructing a multimodal feature extraction network, it can effectively integrate heterogeneous data from multiple sources such as performance evaluation, job duties, skill certification, etc., to realize high-precision identification of the shortcomings of electric power practitioners' competence, whose mathematical expression can be expressed as Equation 2):

$$\hat{Y} = f_{\theta} \left(\sum_{i=1}^n w_i \cdot x_i + b \right) \quad (2)$$

Where $\hat{Y}$ denotes the predicted competency gap value, x_i is the input feature vector, w_i and b are the model weights and bias terms respectively, and f_{θ} denotes the nonlinear mapping function defined by the neural network parameter θ .

3.2 Enhanced learning-driven dynamic path planning

In the personalized training path planning for highly skilled personnel in the electric power industry, reinforcement learning, as a machine learning method that can continuously optimize the decision-making strategy by interacting with the environment, shows remarkable adaptability and flexibility. Compared with traditional rule-based scheduling algorithms, reinforcement learning can dynamically adjust the training content and paths based on

learners' historical behavior and real-time feedback, thus achieving more accurate and efficient personalized recommendations^[6]. Especially in the face of large-scale heterogeneous training tasks, the reinforcement learning model can effectively capture individual ability differences and knowledge mastery state, and then build dynamically optimized learning paths. This process is highly similar to the scheduling problem of deep learning training tasks in GPU clusters, in which a deep reinforcement learning-based scheduling system SCHED2 is proposed in literature [7], which significantly improves the task completion efficiency and resource utilization by modeling the task's sensitivity to GPU resources and the cluster fragmentation state^[8]. In the training scenario of the electric power industry, the learner's ability state, learning progress and resource availability can be modeled as the environment state, and the recommended strategy of the learning path can be used as the action space to optimize the learning effect by designing a reasonable reward function^[9].

To this end, this paper designs a path planning algorithm based on Deep Q-Network (DQN), which centers on constructing a multi-dimensional state representation that reflects the learner's ability portrait. The state vector includes, but is not limited to, the learner's knowledge mastery, skill proficiency, learning preference, historical error rate, and other dimensions, which are derived from the knowledge graph system after fusion of multi-source data. By encoding this information as input states for reinforcement learning, the algorithm can select the optimal next learning content or task module at each decision step. Let the current state of the learner be $s_t \in \mathbb{R}^n$, and the set of optional actions be $a_t \in A$, then the Q function can be expressed as Equation 3):

$$Q(s_t, a_t) = \mathbb{E} \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \mid s_t, a_t \right] \quad (3)$$

where $\gamma \in [0, 1]$ is the discount factor, and r_{t+k} denotes the immediate reward obtained at the $t+k$ th step, which is used to measure the knowledge enhancement effect of the learner after completing a certain task. By iteratively updating the Q network parameters, the system can gradually approach the optimal learning path.

To verify the effectiveness of the method in the cultivation of high-skilled personnel in the electric power industry, we constructed a simulated training environment and introduced several groups of learners with different initial abilities to conduct experiments. Figure 2 shows the process architecture of this path planning algorithm, which contains four core modules: state perception, action selection, environment feedback and strategy update.

In order to improve the efficiency of model training, this paper draws on the literature [10] on the method of multi-expert collaborative labeling of data, and constructs a learning task evaluation standard system jointly formulated by a number of electric power industry experts. By introducing a multi-expert consistency judgment mechanism, the accuracy and generalization ability of the training samples are ensured. The experimental results show that the method has significant advantages in enhancing the speed of learners' knowledge acquisition and reducing the waste of training resources.

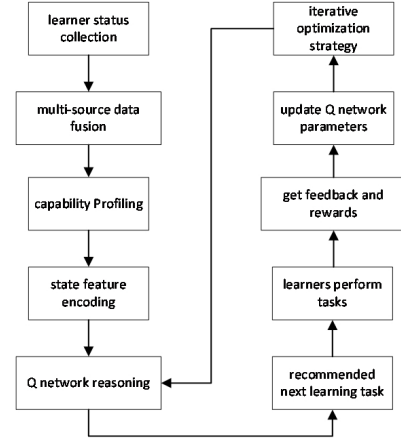


Figure 2: Flow chart of personalized training path planning driven by intensive learning

4 System realization and case validation

4.1 System architecture and functional module design

The design of the system architecture plays a crucial role in the construction of the high-skilled personnel training system in the electric power industry. The system should be based on multi-source heterogeneous data and integrate intelligent technologies such as knowledge graph, deep learning and reinforcement learning to realize the accurate construction of the ability portrait of highly skilled personnel and the intelligent planning of personalized training paths. The overall architecture of the system can be divided into four core modules: data collection layer, knowledge modeling layer, competency analysis layer and training decision-making layer. Among them, the data collection layer is responsible for integrating multi-source information from the simulation training system, skills assessment center and on-site operation data to build a unified data resource pool to provide support for subsequent analysis. The knowledge modeling layer relies on knowledge mapping technology to express professional knowledge such as power equipment structure, operating procedures, fault handling logic, etc. in a graphical structured manner, and to realize the semantic association and reasoning ability of knowledge. The capability analysis layer conducts feature extraction and capability assessment of trainees' behavioral data through deep learning models to form a multi-dimensional capability portrait; the training decision-making layer dynamically adjusts the training path based on reinforcement learning algorithms to enhance the personalization and adaptability of training^[11]. Table 2 shows the structure of the system's functional modules.

4.2 Case validation and analysis in typical scenarios

In the context of the digital transformation of the electric power industry, the construction of competency portraits and training path planning for highly skilled personnel has become a key means

Table 2: The structure of the system’s functional modules.

ModuleHierar- chy	Module Name	Main Function Description	Input Data Type	Output data type	Technical Support
Layer 1	Data Collection Module	Collect multi-source data such as simulation training records, test scores, operation logs, etc.	Structured/un-structured text, logs	Standardized data set after cleaning	Data cleaning, ETL tools
Layer 2	Knowledge modeling module	Construct knowledge graph in power domain, realize semantic association and reasoning at module level	Professional texts, Technical manuals	Domain ontology and relationship mapping	Knowledge extraction, graph database
Layer 3	Competency analysis module	Learner competency assessment and profile construction based on deep learning models	Learner behavior data, test results	Multi-dimensional ability portrait vector	CNN, LSTM, BERT models
Layer 4	Training Decision Module	Generate personalized training paths using reinforcement learning algorithms	Trainee profiling, knowledge graph	Dynamic recommendation of training content	DQN, PPO algorithms
Layer 5	User Interaction Module	Provide visual interface and feedback mechanism to support the monitoring and optimization of the training process.	Teachers’ instructions, trainees’ feedback	Training effect evaluation report	Web front-end, BI tools

to improve the comprehensive quality and job fitness of talents. To verify the practical application effect of the proposed competency portrait model and personalized training path planning method, this paper selects a provincial electric power company as a typical case. It carries out a systematic validation and analysis in combination with the actual operation of its simulation training base^[12]. In this case, through the collection of multiple sources of heterogeneous data such as the trainees’ operation data in the simulation system, behavioral logs of the online learning platform, and performance appraisal results, a competency portrait model based on the knowledge graph was constructed, realizing the multi-dimensional portrayal of the trainees’ skill level, knowledge mastery and operation proficiency. On this basis, deep learning and reinforcement learning algorithms are used to dynamically generate personalized training paths, which are accurately pushed to meet different job requirements and individual differences. Figure 3 shows the implementation process of personalized training path planning, covering the whole process from data collection, image modeling to path generation and feedback optimization.

In the study of the construction of the ability portrait of highly skilled personnel in the electric power industry and AI personalized training path planning, model performance evaluation and comparative analysis are the key links to verify the effectiveness and practicality of the system. To ensure the accuracy of the ability portrait and the adaptability of the training path planning, this paper adopts the multi-source data fusion technology, combines the knowledge graph and deep learning algorithm, and constructs a dynamic optimization model based on reinforcement learning. By training and testing the model under different algorithmic frameworks, we evaluate its comprehensive performance in talent competency identification, skill gap analysis and training path generation

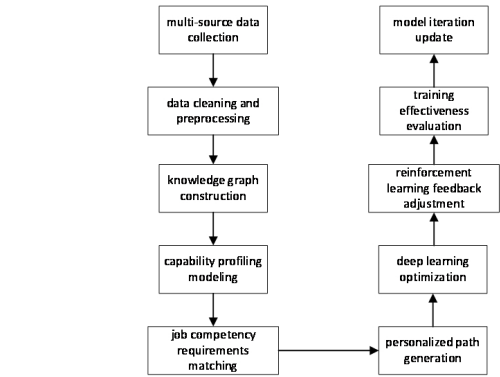


Figure 3: Implementation flow chart of personalized training path planning

to further validate the advantages of the proposed method over traditional training planning methods.

Compared with traditional training, the experimental results are shown in Table 3

The results show that compared with the traditional training methods, the method is significantly optimized in terms of skill attainment rate, training content matching, and employee satisfaction. Training efficiency is improved by about 31%, and trainee satisfaction reaches more than 91%, which significantly improves the training efficiency and employee competence suitability, and verifies its effectiveness.

Table 3: Comparison of experimental results.

Indicators	Traditional training	AI personalized training	Improvement
Skill attainment rate	63%	94%	+31%
Average Achievement Period	4 months	3 months	-25%
Match of training content	42%	75%	+33%
Employee Satisfaction	76%	91%	+15%

5 Conclusion

This study proposes a competency portrait construction method based on multi-source data fusion and an AI-driven personalized training path planning model for the pain points of high-skilled personnel training in the electric power industry. Through empirical verification, the method can effectively solve the traditional training supply and demand mismatch problem, realize the benign cycle of “promoting training with ability and improving ability with training”, and provide technical support for talent training in the digital transformation of enterprises. Future research can further integrate multimodal data perception and natural language processing technology to improve the system’s understanding of trainees’ cognitive state, and at the same time, explore the migratory application of the model in other fields of engineering education and vocational training^[13].

References

- [1] Wang Jing. Thoughts and Suggestions on Digital Transformation of Human Resource Management in Power Grid Enterprises[J]. Chinese Information, 2025 (03): 279-280.
- [2] Wang Zhixuan. Innovative Electric Power Simulation Training Service Skill Talent Cultivation - Speech at the 2011 National Electric Power Industry Simulation Training Work Conference[J]. China Electric Power Education, 2011, (33):2-3.
- [3] LIU Juan, JIN Ming, ZHANG Lianjun. Exploration on the cultivation of top talents with the help of knowledge map[J]. Education and Teaching Forum, 2025, (3):93-96.
- [4] El Asri H., Retbi A., Bennani S. STRATEGIC ADVANCEMENT OF BUILDING INFORMATION MODELLING PRACTICES THROUGH KNOWLEDGE GRAPH CONSTRUCTION USING ONTOLOGY AND DEEP LEARNING[J]. journal of theoretical and applied information technology, 2024, 102(4):1321-1332.
- [5] Zhang W. Optimization of talent cultivation mechanism for vocational undergraduate marketing majors based on knowledge mapping[J]. China Strategic Emerging Industries, 2025, (9):169-171.
- [6] Liu R., Yin G., Liu Z. Tian Y. Reinforcement learning with time intervals for temporal knowledge graph reasoning [J]. Information systems, 2024, 120 (2): 1.1-1.12.
- [7] A. Eliseyev, T. Aksenova. Personalized adaptive instruction design (PAID) for brain-computer interface using reinforcement learning and deep learning: simulated data study [J]. Brain-Computer Interfaces, 2019, 6(2):36-48.
- [8] A. Eliseyev, T. Aksenova. Personalized adaptive instruction design (PAID) for brain-computer interface using reinforcement learning and deep learning: simulated data study[J]. Brain-Computer Interfaces, 2019, 6(2):36-48.
- [9] Xiangdong Zheng, Yuxin Wu, Lianhong ZhangMaobin TangFusheng Zhu. Priority-aware path planning and user scheduling for UAV-mounted MEC networks: a deep reinforcement learning approach [J]. Physical Communication, 2024, 62(2): 102234.1-102234.8
- [10] Yunteng Luan, Xukun Chen, Hanyu Zhao, Zhi Yang, Yafei Dai. SCHED²: Scheduling Deep Learning Training via Deep Reinforcement Learning[C]. IEEE Global Communications Conference.2019.
- [11] Findings from Shanghai University of Engineering Science Provide New Insights into Networks (Application of Emd Combined With Deep Learning and Knowledge Graph In Bearing Fault) [J]. Network Daily News, 2023, (21):20-20.
- [12] Chi Xiujuan. The application of personalized training mode in enterprise talent cultivation[J]. China Management Informatization,2016,019(24):73-74
- [13] Qiu Qinqun, Ma Kai, Lv HairongTao LiufengXie Zhong. Construction and Application of a Knowledge Graph for Iron Deposits Using Text Mining Analytics and a Deep Learning Algorithm [J]. Mathematical geosciences, 2023, 55(3): 423-456.